

a) What will be output of the following programme?

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int array[ 6 ] = { 1, 2, 3, 4, 5, 6 } ;
```

```
    int i ;
```

```
    for ( i = 0 ; i <= 25 ; i++ )
```

```
        printf ( "%d\n", array[ i ] ) ;
```

```
    return 0 ;
```

```
}
```

What will be the output of the following C program? Explain the logic.

```
int main(){  
    char name[]={"National Institute of Technology"};  
    printf("%d",sizeof(name)/sizeof(char));  
    return 0;  
}
```

2. What will be the output of the following C program? Justify your answer.

```
int main(){
    int a[] = {1, 2, 3, 4, 5};
    int sum = 0;
    for(int i = 0; i < 5; i ++){
        if(i % 2 == 0)
            sum += a[i];
        else
            sum -= a[i];
    }
    printf("%d",sum);
    return 0;
}
```

a) What will be the output of the following program?

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int arr[] = { 2, 3, 4, 5 };
```

```
    printf("%d ",arr[2]); printf("%d ",2[arr]); printf("%d ",*(2+arr));
```

```
    printf("%d ",*(arr+2)); return 0;
```

```
}
```

b) Why is it mandatory to specify the number of columns while declaring/initializing multi-dimensional arrays in C? What are Advantages and Disadvantages of Arrays?

2

2+2

2

a) What will be output of the following programme?

```
#include <stdio.h>
int main()
{
    int array[ 6 ] = { 1, 2, 3, 4, 5, 6 };
    int i;
    for ( i = 0 ; i <= 25 ; i++ )
        printf ( "%d\n", array[ i ] );
    return 0 ;
}
```

b) Differentiate between break and continue with an example.

c) Why do we require an Array of Structures? Suppose we have 100 students in a class. How will you store the values of the information of the students? Display the values of all the students